STUDENT NAME: AMANDEEP	MODULE: My Practice	PRACTICE SET 004
BATCH: ABES_JAS26_FSD JAVA_IIIYear_Batch 02	DOMAIN: Gen AI & Prompt Engineering	STATUS: In Progress

Practice Set 4 – Chatbot Dialogue Design

Design conversational system prompts for multi-turn university helpdesks, Socratic tutoring pedagogy, interactive career pathways, and graceful fallback disambiguation.

KEY LEARNING OBJECTIVES:

- Construct stateful multi-turn system prompts for academic and operational support.
- Implement Socratic inquiry prompts that guide learners without prematurely providing direct solutions.
- Architect interactive diagnostic consultation flows that synthesize user input into personalized roadmaps.
- Design graceful disambiguation and fallback strategies for vague or emotional user inputs.

STEP-BY-STEP WORKSHEET INSTRUCTIONS:

Step 1: Define the chatbot's persona, operational boundaries, primary intent, and escalation protocols.
Step 2: Establish the multi-turn conversational state machine (greeting, diagnosis, validation, resolution).
Step 3: Implement strict guardrails (e.g., refusal to hallucinate institutional policy, Socratic questioning limits).
Step 4: Write your engineered system prompt in the workspace and test edge-case handling.

EXERCISE 4.1: MULTI-TURN UNIVERSITY HELPDESK CONVERSATIONAL FLOW

Problem Statement: Engineer a system prompt for 'CampusBot', the automated registrar helpdesk assistant for engineering undergraduates. CampusBot must handle multi-turn inquiries regarding semester course add/drop deadlines, prerequisite waiver petitions, and tuition installment schedules. The prompt must define: Persona (warm, empathetic, highly structured), 3-step verification protocol, clear boundary on grade disputes, and seamless human escalation triggers.

Baseline / Reference Prompt:

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"Act as a college bot and help students with courses and fees."
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STUDENT RESPONSE WORKSPACE:

[1] Role & Persona: _____

[2] Task & Directives: _____

[3] Constraints & Format: _____

[4] Complete Engineered Prompt:

EXERCISE 4.2: SOCRATIC STEM STUDY ASSISTANT & DIALOGUE SCAFFOLDING

Problem Statement: Design a system prompt for 'MentorSigma', a specialized Socratic programming and algorithms tutor. When a student presents a broken Dynamic Programming algorithm or asks for the direct solution, MentorSigma must NEVER provide raw code or direct answers. Instead, it must diagnose the student's conceptual block, ask exactly ONE targeted guiding question per turn, and validate progress incrementally.

Baseline / Reference Prompt:
"You are a programming tutor. Help me solve this dynamic programming problem without giving away the answer right away."

STUDENT RESPONSE WORKSPACE:
[1] Role & Persona: _____
[2] Task & Directives: _____
[3] Constraints & Format: _____
[4] Complete Engineered Prompt:

EXERCISE 4.3: INTERACTIVE CAREER GUIDANCE & DIAGNOSTIC PATHWAY ADVISOR

Problem Statement: Develop a conversational system prompt for 'Pathfinder AI', a career consultation chatbot for CS undergraduates. The bot must lead a structured 3-stage consultation: Stage 1 (Technical Strengths & Project Experience), Stage 2 (Industry Interests & Work-Style Preferences), and Stage 3 (Generating an actionable 18-month roadmap with certifications, portfolio projects, and interview preparation targets).

Baseline / Reference Prompt:
"Help me figure out what career in tech I should do by asking me some questions."

STUDENT RESPONSE WORKSPACE:
[1] Role & Persona: _____
[2] Task & Directives: _____
[3] Constraints & Format: _____
[4] Complete Engineered Prompt:

EXERCISE 4.4: DISAMBIGUATION & FALLBACK FOR UNCLEAR / INCOMPLETE USER REQUESTS

Problem Statement: Build a system prompt instructing a support chatbot to handle ambiguous, emotionally charged, or underspecified queries (e.g., 'My portal is broken and nothing works!' or 'I need a refund right now'). The prompt must enforce the 'Acknowledge - Categorize - Disambiguate' framework: de-escalating frustration, presenting the 3 most probable service categories with clickable option letters, and providing a fallback escalation path.

Baseline / Reference Prompt:
"Respond to users who ask vague or confusing questions so they don't get frustrated."

STUDENT RESPONSE WORKSPACE:
[1] Role & Persona: _____
[2] Task & Directives: _____
[3] Constraints & Format: _____
[4] Complete Engineered Prompt:

SELF-REVIEW CHECKLIST:

- [ ] **Persona Consistency:** Conversational identity and boundaries maintained across multi-turn exchanges.
- [ ] **Pedagogical Scaffolding:** Guides discovery without immediate answer dumping or code leakage.
- [ ] **Turn-Taking & Brevity:** Asks one focused question at a time to prevent cognitive overload.
- [ ] **Ambiguity Handling:** Gracefully de-escalates emotional queries and offers structured disambiguation options.
- [ ] **Escalation Protocol:** Clear triggers defined to hand off complex or sensitive cases to human staff.

CRITICAL REFLECTION & LEARNING SYNTHESIS:

- *Why is enforcing 'one question per turn' in chatbot prompts critical for maintaining productive user dialogues?*
- *How does the Socratic dialogue pattern transform an AI from a passive answer engine into an active learning partner?*

Reflection Notes:

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_____

COMPLETION STATUS:

[ ] COMPLETED (Meets All Rubric Criteria)
[ ] NEEDS REVISION (Resubmit Refined Prompts)

Evaluator Signature: _____

EVALUATOR & MENTOR REMARKS:

Feedback Notes:
_____
Rubric Evaluation: [ ] Clarity [ ] Constraints [ ] Edge Handling